Name_____ Date_____ Period_____

LESSON 4-5 · HOMEWORK PACKET · SHEET 1 OF 2

Find the Whole

Maryland Standard 6.AT.4 · Learning target – I can find the whole when I know a part and the percent it represents.

REMEMBER

- The **part** is what you know. The **whole** is 100%.
- Step **down** to a friendly percent, then build **up** to 100%.
- Percent below 100%? The whole is **greater** than the part.

WORKED EXAMPLE

24 is 40% of what number?

Step 1 40% is 24, so 10% is 24 ÷ 4 = 6.

Step 2 100% is 10 × 6 = 60.

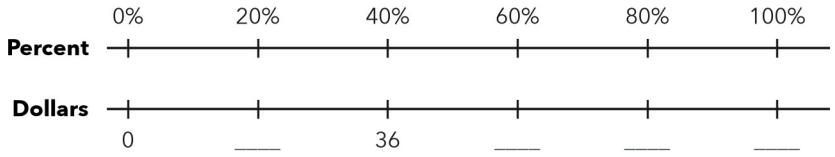
Step 3 Check: 0.4 × 60 = 24, so it checks.

Answer 60

PRACTICE

show your work · label units

1. Bike Helmet. Andre has saved $36. That is 40% of the cost of a bike helmet. Fill in the double number line. Then find the cost of the helmet.



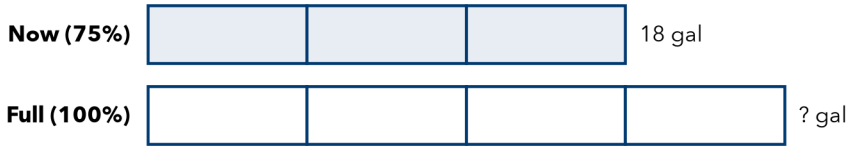
Cost_____ dollars

2. Find each whole.

a. 13 is 10% of _____ **b.** 11 is 25% of _____ **c.** 21 is 30% of _____

d. 32 is 40% of _____ **e.** 7 is 5% of _____ **f.** 54 is 60% of _____

3. Water Tank. A water tank is 75% full. It holds 18 gallons now. Each part of the tape diagram is 25% of the full tank. How many gallons does the tank hold when it is full?



Answer_____ gallons

4. Chapter Book. Lena has read 120 pages. That is 75% of her book. How many pages long is it?

Answer_____ pages

5. Reason. 20 is 40% of number A. 20 is also 80% of number B. Without solving, which number is greater, A or B? Explain. Then find both numbers.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Tell someone at home: if 10 is 50% of a number, what is the number? Explain how you know. **Sentence frame:** "I know the number is ___ because ___."

Name_____ Date_____ Period_____

LESSON 4-5 · HOMEWORK PACKET · SHEET 2 OF 2

Solve for the Whole

Maryland Standard 6.AT.4 · Learning target – I can write and solve an equation to find the whole from a part and a percent.

REMEMBER

- Write **part = percent × whole**, with *w* for the whole.
- Solve by dividing: *w* = part ÷ decimal.
- **Check:** decimal × whole should give the part.

WORKED EXAMPLE

$18 is 60% of a shirt's first price. Find the first price.

Step 1 Write the equation: 0.6 × *w* = 18

Step 2 Divide: *w* = 18 ÷ 0.6 = 180 ÷ 6 = 30

Step 3 Check: 0.6 × 30 = 18, so it checks.

Answer **$30**

PRACTICE

show your work · label units

1. Solve each equation for *w*.

a. 0.2 × *w* = 14 ____ **b.** 0.25 × *w* = 12 ____ **c.** 0.8 × *w* = 64 ____

2. 33 is 55% of what number? Write an equation and solve it.

a. Equation: _____

b. *w* = _____

3. **Field Trip.** A class has collected $315 for a trip. That is 75% of the total cost. Find the total cost. How much more does the class need?

Show your work.

Total and still needed_____

4. **Savings Goal.** Tasha has saved $42. That is 35% of her goal. Write an equation and find her goal.

Equation and goal_____

5. **Find the Error.** Jamal solved "28 is 35% of *w*" as *w* = 28 ÷ 35 = 0.8. Explain his mistake. Solve and check.

_____

_____

TEST PRACTICE

choose the best answer

6. 45 is 30% of a number *w*. Which equation can you use to find *w*?

A 0.3 × 45 = *w* **B** 0.3 × *w* = 45 **C** 45 × *w* = 30 **D** *w* ÷ 45 = 0.3

7. A team won 18 games. That is 60% of its games. How many games did it play?

A 10.8 **B** 24 **C** 30 **D** 78

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Explain to someone at home why the whole must be bigger than the part when the percent is less than 100%.